

INTERNSHIP REPORT
ON
AI VOICE AGENTS, ERP AND CRM SYSTEMS

By

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SRINIVAS INSTITUTE OF TECHNOLOGY

(Accredited by NAAC with “A” grade)

MANGALURU-574143, KARNATAKA



Department of Artificial Intelligence and Machine Learning

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi-590018

2025-2026

SRINIVAS INSTITUTE OF TECHNOLOGY
VALACHIL, MANGALURU - 574 143, KARNATAKA

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

It is to certify that the report on Industrial Internship carried out at

Rooman Technologies Pvt. Ltd.

is a Bonafide work carried out by

PRANEETH P K

4SN22AI047

in partial fulfilment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2025 - 26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the Departmental library. The Internship report has been approved as it satisfies the academic requirements in respect of Industrial Internship prescribed for the Bachelor of Engineering Degree.

Prof. Parvathraj K M M

Mr. Sandeep Singh

Prof. Daya Naik

Dr. Shrinivasa Mayya D

Internal guide

External Guide

Head of the Department

Principal

Examiners:

Name

Signature

1.

2.

APPENDIX – I

CONSENT LETTER OF INTERNSHIP



SRINIVAS INSTITUTE OF TECHNOLOGY

(ENGINEERING COLLEGE OF SRINIVAS GROUP)

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Ref. No. SIT/INTERN/2025-26/347

OFFICE OF THE PRINCIPAL

Date: 26/12/2025

To,

The HR,
Rooman Technologies Pvt. Ltd.,
#30,12th Main, 1st Stage,
Rajajinagar, Bengaluru – 560010.

Sub: Request to grant permission to undergo internship training at your organization.

Greetings from Srinivas Institute of Technology!

As a part of academic requirement, the students of 4th year B.E in Artificial Intelligence and Machine Learning Engineering branches are required to take up the internship or project work, to enhance their practical knowledge, skills and also to be acquainted with the latest technologies and tools. In this regard the below mentioned student of our college is interested to take up the internship at your esteemed organization for a period of fifteen weeks.

Sl.No	NAME	USN
1.	PRANEETH P K	4SN22AI047

We kindly request you to permit him to take up the training in your organization.

Thanking you.

Yours faithfully,


Dr. Shrinivasa Mayya D
PRINCIPAL
PB: OC4



APPENDIX - II

INTERNSHIP COMPLETION CERTIFICATE



INTERNSHIP COMPLETION CERTIFICATE

Date: 09-05-2026

Respected Sir/Madam ,

This is to certify that **Mr.. Praneeth P K, S/O Pushpalatha**, bearing USN **4SN22AI047**, from **Srinivas Institute of Technology**, has successfully carried out and completed the **On-Site** internship in the domain of **AI Product Developer** at Rooman Technologies from **21-01-2026** to **06-05-2026**.

During the internship period, the student demonstrated regularity, punctuality, sincerity, and strong interest towards learning and skill development. The student actively participated in the assigned tasks, gained practical work experience, and successfully fulfilled the internship objectives assigned by the organization.

Based on the overall performance, dedication, and commitment shown throughout the internship period, a score of **100** out of 100 is awarded.



Name: Sandeep Singh

Designation: AI Ops Engineer, Rooman Technologies

Organization: Rooman Technologies

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ACKNOWLEDGEMENT

The success and final outcome of this internship required a lot of guidance and assistance from many people, and I am extremely fortunate to have got their support all along the completion of my internship.

I am highly grateful and would like to express my wholehearted thankfulness to my Internship Guide **Dr. Parvathraj K M M, Professor**, who has been my source of inspiration. He has communicated various ideas for improving the internship project and has been especially enthusiastic in giving his opinions and critical reviews in a constructive manner. I will remember his contribution forever.

I express my gratitude to my Internship Coordinator **Prof. Madhusudhan S, Assistant Professor**, for his constant support and motivation.

I express my gratitude to **Rooman Technologies Pvt. Ltd.** for providing me the opportunity to do my internship. I am especially grateful to **Mr. Sandeep Singh** for his guidance and support throughout the internship.

I express my deep sense of gratitude to **Prof. Daya Naik, Head of the Department**, for her cordial support, valuable information and guidance, which helped me in completing this internship through various stages.

I also express my heartfelt gratitude to Principal **Dr. Shrinivasa Mayya D**, for his kind co-operation and encouragement which helped in the completion of my internship.

I am thankful to and fortunate enough to get constant encouragement, support and guidance from all the **Teaching and Non-teaching staff** of the Department of Artificial Intelligence and Machine Learning who helped in successfully completing my internship.

Lastly, I would like to thank my **Parents** for their moral support and my **Friends** with whom I shared my day-to-day experiences and received lots of suggestions that improved my quality of work.

Praneeth P K
4SN22AI047

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CHAPTER 1

ABOUT THE ORGANIZATION

1.1 INTRODUCTION TO THE ORGANIZATION

Rooman Technologies is a technology and skill development organization headquartered in Bengaluru, Karnataka, India. The organization focuses on providing industry-oriented technical education, internship opportunities, and practical learning experiences for students and professionals in emerging technology domains.

The company was established with the objective of bridging the gap between academic learning and industry requirements through hands-on training, project-based learning, and real-world technology exposure. Rooman Technologies provides opportunities for students to improve their technical knowledge, practical implementation skills, and professional capabilities through internships, workshops, and technology-driven programs.

The organization offers learning and development programs in domains such as Artificial Intelligence, Machine Learning, Cloud Computing, Cybersecurity, Full Stack Development, Data Analytics, and Software Development. Along with technical training, the organization also encourages innovation, teamwork, problem-solving, and continuous learning through practical activities and project development environments.

Rooman Technologies focuses on helping learners become industry-ready by combining theoretical concepts with practical implementation and real-time exposure to modern technologies. The organization provides a professional and collaborative environment that supports skill enhancement, career development, and technology-based learning initiatives.

The company has collaborated with academic institutions, industry experts, and technology partners to support employability and technical skill development programs across different regions. Through its internship and development activities, the organization continues to contribute toward building skilled professionals capable of adapting to modern industry standards and technological advancements.



Figure 1.1.1: Rooman Technologies Pvt. Ltd. Logo

1.2 COMPANY OVERVIEW

Rooman Technologies is a technology and skill development organization headquartered in Bengaluru, Karnataka, India. The company was established in the year 1999 with the objective of providing industry-oriented technical education, practical learning opportunities, and technology-based development programs for students and professionals.

The Founder and CEO of the organization is Manish Kumar, who has played a significant role in building the organization's presence in technical education, internship programs, and technology learning initiatives.



Figure 1.2.1: CEO of Rooman Technologies Pvt. Ltd.

Over the years, Rooman Technologies has expanded its presence in various technology and learning domains by collaborating with academic institutions, industry experts, and technology partners. The organization focuses on bridging the gap between academic knowledge and industry requirements through practical implementation, internships, and project-based learning approaches. The company supports learning and development activities in emerging technology domains and

encourages students to improve their technical, analytical, and professional skills through hands-on exposure and real-world project environments.

Rooman Technologies also promotes innovation, teamwork, continuous learning, and practical implementation of technologies that help learners understand modern industry standards and development practices.

Apart from training and internship programs, Rooman Technologies is also involved in software product development and enterprise technology solutions. The organization works on internal systems such as ERP platforms, CRM solutions, AI-based applications, automation systems, and real-time development environments that support operational workflows, communication systems, and business processes. The company also focuses on emerging technologies including Artificial Intelligence, cloud technologies, automation tools, and AI-assisted development systems to support modern digital transformation initiatives.

1.3 VISION

The Vision of Rooman Technologies is to integrate global innovation, technology, and skill development to empower people, society, and businesses. The organization focuses on creating a technology-driven learning ecosystem that supports professional growth, innovation, and industry-oriented skill enhancement.

The Vision: “To empower people, society, and businesses through global innovation, modern technology, and industry-oriented skill development.”

1.4 MISSION

Rooman Technologies aims to contribute toward developing skilled professionals who can adapt to modern technological advancements and support the growth of industries and society through knowledge, creativity, and practical implementation.

The Mission: “Impart quality training for empowerment of youth to make India skill capital of the world. Integrate Global Technologies to introduce innovative Products and Solutions. Increase global presence through Associations, Collaborations and Partnerships. To setup Global Education Campus housing top universities of the world.”

The organization also promotes practical learning, innovation, teamwork, and professional skill development through internships, workshops, certification programs, and real-world project exposure.

1.5 SERVICES OFFERED

Rooman Technologies provides various technology-focused services and learning programs aimed at improving technical skills, employability, and practical industry exposure for students and professionals. The organization offers industry-oriented training, internship opportunities, certification programs, and technology-based learning initiatives in emerging technical domains.

The company focuses on practical implementation, project-based learning, and real-world technology exposure through workshops, live projects, internships, and development activities. Rooman Technologies also collaborates with academic institutions, industry experts, and technology partner to support technical education and skill development initiatives.

1.5.1 Training and Skill Development

Rooman Technologies provides industry-oriented technical training programs in various emerging technology domains. The organization offers classroom, online, and hybrid learning programs designed to improve both theoretical understanding and practical implementation skills.

The company provides training in areas such as:

- Artificial Intelligence and Machine Learning
- Cloud Computing and DevOps
- Full Stack Development
- Cybersecurity and Networking
- Data Analytics
- Python and Java Programming

The training programs include practical exercises, lab sessions, real-world projects, and industry-relevant learning methodologies that help students improve technical and analytical skills.

1.5.2 INTERNSHIP PROGRAMS

Rooman Technologies conducts internship programs that provide students with practical exposure to modern technologies, development environments, and real-world project workflows. The internship programs focus on hands-on implementation, project development, teamwork, and industry-oriented learning approaches.

The organization offers internships in domains such as:

- AI and Machine Learning
- DevOps and Cloud Computing
- Full Stack Development
- Cybersecurity
- Data Analytics

The internship structure includes mentorship, project-based learning, technical guidance, and professional development activities that help students gain practical industry experience.

1.5.3 TECHNOLOGY SOLUTIONS AND DEVELOPMENT

Apart from training and internships, Rooman Technologies also supports technology-based development activities and innovation-focused initiatives. The organization promotes practical implementation of technologies through software projects, automation systems, cloud-based applications, and AI-driven solutions.

The company encourages learners and interns to work on real-time development activities that improve understanding of software development workflows, project implementation, and modern technology practices. The organization also integrates emerging technologies and AI-enabled approaches into learning and development activities.

1.5.4 Career Support Services

Rooman Technologies provides career support and professional development services to help students improve employability and industry readiness. The organization supports learners through mentorship, interview preparation, technical assessments, resume-building activities, and placement assistance.

The company also conducts placement drives, mock interviews, aptitude preparation sessions, and career guidance activities that help students improve communication skills, confidence, and capabilities.

1.6 TECHNOLOGIES USED IN THE ORGANIZATION

Rooman Technologies works with various modern technologies, development platforms, AI tools, and cloud-based systems to support technical learning, software development, internships, and industry-oriented solution implementation. Along with technical education and training programs, the organization has also expanded into corporate training, IT services, software development, and technology-driven solution development for enterprises and institutions.

The organization focuses on emerging technology domains such as Artificial Intelligence, Machine Learning, Cloud Computing, DevOps, Cybersecurity, Data Analytics, Software Development, and Full Stack Development. Rooman Technologies also integrates AI-enabled learning approaches, automation systems, and modern development environments into its training and development ecosystem.

The company encourages practical implementation and project-based learning through software projects, AI-driven applications, cloud platforms, automation tools, and real-world development activities. Interns and learners are provided exposure to modern development workflows, collaborative environments, and technology-oriented project implementation practices.

Some of the major technologies and tools used in the organization include:

- Python
- Java
- JavaScript
- HTML and CSS
- React.js
- Node.js

- SQL and Database Technologies
- Cloud Computing Platforms (AWS, Azure)
- Artificial Intelligence and Machine Learning Tools
- APIs and Automation Tools
- AI-Assisted Development Tools
- No-Code and Low-Code Development Platforms
- DevOps and Deployment Tools

Rooman Technologies has also partnered with academic institutions, government initiatives, and industry organizations to support technical education, innovation, and enterprise technology solutions. The organization collaborates with partners such as NSDC, NASSCOM, Skill India, Cisco, Microsoft, Red Hat, and other technology and certification providers to enhance learning opportunities and industry exposure for students and professionals

1.7 WORK ENVIRONMENT AND CULTURE

The work environment at Rooman Technologies is professional, technology-driven, and collaborative. The organization follows an industry-oriented approach where interns and team members are encouraged to work on real-time development activities, live projects, and client-oriented tasks.

The organization follows a collaborative and development-oriented work culture where interns and developers participate in project discussions, implementation activities, debugging processes, sprint reviews, testing workflows, and documentation tasks. Exposure to live projects and production-oriented systems helped in understanding real-time software development practices and professional engineering workflows followed in technology environments.



Figure 1.7.1: Campus of Rومان Technologies Pvt. Ltd.

During the internship period, interns were provided exposure to practical software development workflows, AI-based solutions, automation systems, and modern development practices. The organization promotes teamwork, innovation, problem-solving, and continuous learning through hands-on project implementation and collaborative development environments.

The company provides opportunities for interns to work with emerging technologies, AI-assisted development tools, cloud platforms, APIs, and modern software frameworks. The work culture encourages active participation in project discussions, development processes, testing activities, and solution implementation approaches followed in real-world industry environments.

Rومان Technologies maintains a supportive and growth-oriented culture where mentors and team members provide guidance throughout project development activities. The organization focuses on helping interns improve technical knowledge, professional communication, project handling abilities, and practical implementation skills through real-time industry exposure.

1.8 ROLE OF THE ORGANIZATION IN INTERNSHIP

During the internship period, Rومان Technologies played an important role in providing practical industry exposure, technical guidance, and real-world development experience. The organization provided opportunities to work on live projects, client-oriented tasks, and AI-based development activities that helped in understanding modern software development workflows and industry practices.

The internship was focused on practical implementation and product development activities rather than only theoretical learning. Interns were encouraged to participate in project discussions, development processes, testing activities, and solution implementation using modern technologies and AI-assisted development tools.

The organization provided mentorship and technical support throughout the internship period, helping interns improve problem-solving abilities, technical implementation skills, teamwork, communication, and project handling experience. Exposure to real-time project environments helped in understanding development methodologies, collaborative workflows, and practical industry standards followed in professional technology environments.

Through this internship, the organization contributed significantly toward improving technical confidence, professional understanding, and practical knowledge in AI product development and software technologies.

1.9 ACHIEVEMENTS AND STRENGTHS

Rooman Technologies has established a strong presence in the fields of technical education, internships, skill development, and technology-oriented learning initiatives. The organization has received several recognitions and awards for its contribution toward industry-oriented training, professional skill development, and technology education initiatives.

Some of the notable awards and recognitions received by the organization include:

- SKOCH AWARD - India's Honest Independent Award
- Prime Time Global Education Excellence Awards-2014
- NSDC - Best Practice Recognition Award
- ASSOCHAM - Summit-cum-Awards in Skilling India
- All India Technical and Management Council - Leading Skill Training Partner Award
- Brands Academy - Education Excellence Award
- IT entrepreneurs' meet in China

The organization has also participated in AI summits, technology showcases, innovation events, and industry collaboration activities related to emerging technologies and modern development practices. These achievements reflect the organization's commitment toward innovation, practical learning, and technology-driven development initiatives.

1.10 CONCLUSION

Rooman Technologies played an important role in providing practical exposure, technical guidance, and professional learning opportunities during the internship period. The organization provided a professional and technology-driven environment where interns could work on live projects, client-oriented tasks, and real-world development activities.

The internship experience helped in improving technical knowledge, project understanding, teamwork, communication skills, and practical implementation abilities. Exposure to modern technologies, AI-based development practices, and collaborative project environments contributed significantly toward professional and technical growth.

Overall, the internship at Rooman Technologies provided valuable industry experience and helped in understanding real-world software development workflows, modern technologies, and professional work culture.

CHAPTER 2

OBJECTIVES OF INTERNSHIP

2.1 INTERNSHIP OBJECTIVES

The primary objective of the internship at Rooman Technologies was to gain practical industry exposure and understand real-world software development workflows in the field of AI product development. The internship focused on bridging the gap between academic learning and professional industry practices through hands-on implementation and live project experience.

The internship also focused on understanding real-time development environments, collaborative workflows, software implementation methodologies, and practical engineering processes followed in professional technology organizations. Exposure to AI systems, automation workflows, APIs, enterprise applications, and development activities contributed toward improving practical industry understanding and technical capabilities.

Another important objective of the internship was to gain experience with modern tools, platforms, AI/ML technologies, cloud environments, APIs, and collaborative development workflows used in current industry practices. The internship also encouraged participation in project discussions, reviews, evaluations, and real-world problem-solving activities to improve professional confidence and technical capabilities.

2.2 PROFESSIONAL OBJECTIVES

The professional objective of the internship was to develop industry-relevant technical and professional skills required for working in modern technology-driven organizations. The internship focused on improving understanding of project workflows, client-oriented development practices, documentation standards, collaborative development processes, and professional work culture.

- Improving teamwork and collaboration skills
- Gain practical exposure to live projects and real-world implementation
- Understand professional communication and teamwork practices

- Learn modern development tools, AI technologies, and collaborative workflows
- Enhance project handling, documentation, and implementation skills
- Develop confidence in working within professional development environments

Through practical implementation, mentorship, and continuous learning activities, the internship contributed toward improving technical expertise and career readiness in the field of AI product development and software technologies.

2.3 ROLES AND RESPONSIBILITIES

During the internship period at Rooman Technologies, the role assigned was AI Product Developer Intern under the on-site internship program. The internship involved participation in live, industry-oriented projects and real-world development activities aligned with the internship domain.

The major roles and responsibilities during the internship included:

- Working on live and industry-oriented development projects
- Assisting in designing, developing, testing, and documenting assigned tasks
- Collaborating with mentors, team leads, and peers in professional environments
- Participating in daily activities, sprint discussions, reviews, and evaluations
- Applying theoretical knowledge to practical real-world problem solving
- Maintaining documentation, reports, and project-related activities
- Following organizational processes, timelines, and quality standards
- Working with modern tools, AI technologies, cloud platforms, and collaborative development systems

The internship provided practical exposure to real-world software development workflows, project implementation methodologies, and professional development practices followed in modern technology environments

2.4 EXPECTED OUTCOMES OF THE INTERNSHIP

The internship was expected to provide practical industry exposure and improve understanding of real-world software development workflows, AI product development activities, and collaborative engineering environments. The internship aimed to strengthen technical knowledge, analytical thinking, problem-solving abilities, and practical implementation skills through participation in live projects and real-time development activities.

The expected outcomes of the internship included:

- Improved understanding of AI-based systems and software development workflows
- Exposure to live projects, collaborative development practices, and implementation processes
- Enhancement of technical, analytical, and professional communication skills
- Better understanding of project workflows, debugging, testing, and documentation practices
- Practical exposure to modern development tools, APIs, cloud platforms, and AI technologies
- Increased confidence in handling technical tasks, project responsibilities, and real-world development activities.

Overall, the internship was expected to contribute toward professional growth, technical improvement, and career readiness in the field of Artificial Intelligence and software product development.

CHAPTER 3

LEARNING EXPERIENCES

3.1 RESEARCH AND DEVELOPMENT IN AI SYSTEMS

The initial phase of the internship focused on research and development activities related to Artificial Intelligence systems, natural language processing, and conversational AI technologies. The primary objective during this phase was to understand real-world AI workflows, evaluate different technologies, and explore practical implementation approaches used in modern AI-driven systems.

The research activities began with experimentation on Personally Identifiable Information (PII) extraction systems intended for AI voice agent environments. Various Natural Language Processing (NLP) and Named Entity Recognition (NER) approaches were studied and tested to understand their ability to detect and process sensitive information from conversational inputs. Multiple frameworks and libraries such as spaCy, Flair, HuggingFace Transformers, GLiNER, Stanford CoreNLP, and other entity recognition tools were explored during the experimentation process.

Different models and extraction approaches produced varying results depending on the type of input, contextual complexity, response structure, and processing requirements. Some models performed effectively in structured extraction tasks, while others provided better contextual understanding in conversational scenarios. Through continuous experimentation and evaluation, a hybrid implementation approach combining regex-based extraction patterns with LLM-assisted contextual processing was explored to improve extraction reliability and flexibility.

Following the PII extraction research phase, the focus shifted toward AI voice systems involving Text-to-Speech (TTS), Speech-to-Text (STT), and Large Language Models (LLMs). The experimentation process involved understanding real-time conversational AI workflows, streaming architectures, response latency, voice synthesis quality, transcription accuracy, and contextual response generation. Multiple AI models, frameworks, and communication systems were studied to understand their behavior under different real-time conversational conditions.

Further research activities included experimentation with multi-agent voice systems and AI communication frameworks such as LiveKit, Pipecat, and TEN Framework. These activities involved

studying conversational routing workflows, multi-agent coordination, audio session handling, real-time streaming behavior, and dynamic persona-based interactions within AI voice environments.

The research and development phase provided practical exposure to emerging AI technologies, real-time communication systems, model evaluation workflows, and modern AI engineering practices. It also helped in developing analytical thinking, experimentation methodologies, debugging approaches, and practical understanding of real-world AI product development environments.

3.2 AI VOICE AGENT RESEARCH AND EXPERIMENTATION

The next phase of the internship focused on research, experimentation, and implementation activities related to AI voice agent systems. The primary objective during this phase was to understand the architecture and workflow of real-time conversational AI systems involving Text-to-Speech (TTS), Speech-to-Text (STT), and Large Language Models (LLMs).

The experimentation process involved studying different AI voice technologies, real-time communication workflows, and streaming-based conversational systems. Multiple TTS and STT providers, open-source models, and AI frameworks were explored to analyze their performance under different conversational scenarios. Factors such as response latency, voice naturalness, transcription accuracy, contextual understanding, streaming quality, and interruption handling were continuously evaluated throughout the research activities.

Various AI models and communication frameworks demonstrated different strengths and limitations depending on the use case and real-time conditions. Some models performed effectively in low-latency response generation, while others provided better conversational flow, contextual understanding, or voice quality. This experimentation process helped in understanding the trade-offs between performance, scalability, response quality, and infrastructure requirements involved in production-oriented AI voice systems.

The internship also involved experimentation with multi-agent conversational workflows and real-time communication frameworks such as LiveKit, Pipecat, and TEN Framework. These frameworks were explored to understand audio transport, session management, conversational routing, persona switching, and continuous voice interaction handling in AI-driven communication environments.

Further activities included studying multi-persona conversational systems where multiple AI agents

interact dynamically based on user intent and workflow conditions. Testing activities were carried out to analyze agent switching behavior, conversational continuity, synchronization challenges, and streaming stability across real-time voice interactions.

The experimentation phase also provided exposure to AI model integration workflows, voice processing pipelines, real-time debugging approaches, and practical AI system implementation methodologies. Through continuous testing, analysis, and iterative improvements, a deeper understanding of conversational AI engineering, streaming architectures, and production-oriented voice agent systems was developed during the internship period.

3.3 WEEKLY DEVELOPMENT ACTIVITIES

The internship activities were carried out through weekly development tasks, research activities, implementation workflows, debugging processes, and collaborative project contributions. The weekly activities included AI research, voice agent experimentation, ERP and CRM development, Agentic AI Orchestration for Business, workflow integration, testing, and practical implementation of real-world software development practices. The following sections summarize the major activities and learning experiences carried out during each phase of the internship period.

Week 1 – PII Extraction Research and Initial NLP Experimentation

- Studied Personally Identifiable Information (PII) extraction systems for AI voice agent applications
- Performed research and testing on NLP and Named Entity Recognition (NER) approaches
- Experimented with tools and frameworks such as spaCy, Flair, HuggingFace Transformers, GLiNER, Stanford CoreNLP, and other PII extraction libraries
- Analyzed model accuracy, contextual understanding, and extraction performance under different scenarios
- Explored regex-based extraction methods and compared them with AI/LLM-based approaches
- Began developing a hybrid extraction strategy combining regex patterns with LLM-based contextual processing

Week 2 – Advanced PII Workflow Research and AI Model Evaluation

- Continued experimentation with PII detection and structured information extraction workflows
- Tested multiple extraction patterns for emails, phone numbers, IDs, addresses, and sensitive entities
- Evaluated model behaviour under noisy and unstructured conversational inputs
- Improved extraction reliability using layered validation and filtering techniques
- Studied limitations of traditional NER systems in conversational AI workflows
- Worked on improving contextual extraction approaches using AI-assisted reasoning systems

Week 3 – Voice AI Pipeline Research and Technology Exploration

- Started research on AI voice agent systems involving Text-to-Speech (TTS), Speech-to-Text (STT), and Large Language Models (LLMs)
- Explored real-time conversational AI workflows and audio pipeline architectures
- Performed comparative analysis of multiple TTS and STT providers and models
- Studied response latency, transcription accuracy, voice naturalness, and streaming performance
- Tested different LLMs for conversational flow, intent handling, and contextual responses
- Understood challenges involved in low-latency real-time voice communication systems

Week 4 – AI Voice Agent Experimentation and Multi-Agent Workflow Research

- Began experimentation with multi-agent conversational systems and routing workflows
- Explored frameworks and tools such as LiveKit, Pipecat, and TEN Framework for voice agent communication
- Worked on agent coordination and conversational handoff mechanisms

- Tested voice interaction workflows across multiple AI personas and routing conditions
- Analyzed real-time streaming challenges, synchronization issues, and conversational delays
- Studied audio session management and continuous conversational flow handling

Week 5 – Multi-Voice Agent System Development and Workflow Testing

- Continued development and experimentation on multi-voice AI agent systems
- Worked on conversational flow handling and dynamic response routing mechanisms
- Performed testing on TTS/STT model combinations under different real-time conditions
- Evaluated model behavior for latency, interruption handling, and voice continuity
- Debugged communication flow issues and synchronization-related problems in voice pipelines
- Improved agent interaction handling and response generation workflow

Week 6 – ERP System Understanding and Development Environment Setup

- Started working on Rooman ERP development workflows and internal system architecture
- Understood backend/frontend project structure and operational workflows of the ERP system
- Studied module relationships, role-based workflows, and system implementation logic
- Participated in development discussions and workflow analysis activities
- Explored existing modules and implementation approaches used in production systems
- Understood collaborative development practices followed within the organization

Week 7 – Referral SaaS Module Planning and ERP Workflow Development

- Worked on referral-based SaaS workflows integrated with ERP-related systems
 - Participated in planning and development of referral lifecycle implementation
 - Contributed to frontend and backend workflow integration activities
-

- Worked on QR-based referral concepts and public workflow handling
- Participated in implementation planning for user onboarding and lead generation workflows
- Performed testing and validation of workflow logic and integration activities

Week 8 – ERP Module Enhancements and UI Development Activities

- Worked on ERP feature enhancements and implementation-related tasks
- Contributed to frontend interface improvements and workflow refinements
- Participated in debugging and fixing module-related issues
- Worked on dashboard improvements, workflow validation, and usability enhancements
- Assisted in integration activities between frontend and backend systems
- Participated in testing and implementation reviews for ERP workflows

Week 9 – Test Module and Question Bank Development Activities

- Participated in development activities related to test modules and question bank systems
- Worked on structured data handling and CRUD-based workflow implementations
- Assisted in improving admin-to-user workflow interactions and interface behavior
- Worked on validation logic, filtering mechanisms, and UI refinements
- Participated in implementation testing and debugging activities
- Improved workflow consistency and module interaction handling

Week 10 – CRM System Introduction and Leads Workflow Development

- Started working on CRM-based workflows and operational modules
- Understood leads management systems and lifecycle-based workflow implementation
- Worked on data handling, workflow structuring, and CRM interface functionalities

- Participated in frontend/backend integration and API-related activities
- Assisted in testing CRM workflows and operational functionalities
- Explored collaborative workflow handling within CRM modules

Week 11 – CRM Calls Module and Analytics Workflow Development

- Contributed to CRM calls-related workflows and interaction management systems
- Worked on reusable components, filtering systems, and analytics-related implementations
- Participated in dashboard and workflow enhancement activities
- Assisted in debugging and improving CRM operational functionalities
- Worked on API integration and workflow optimization tasks
- Participated in collaborative implementation and testing activities

Week 12 – Advanced CRM Enhancements and Workflow Optimization

- Continued enhancement of CRM modules and operational workflows
- Worked on improving usability, interface consistency, and workflow efficiency
- Assisted in integration-related activities and system behavior validation
- Participated in debugging, issue tracking, and workflow optimization
- Improved module coordination and frontend/backend synchronization
- Participated in review and testing activities for production-oriented workflows

Week 13 – AI Product Development Learning and Personal Project Exploration

- Applied internship learning toward personal AI product development activities
- Worked on experimental AI workflows inspired by real-world implementation practices
- Continued research on AI-assisted communication systems and multi-agent architectures

- Explored conversational AI integration and automation-oriented workflows
- Studied scalable AI product implementation concepts and workflow structures
- Participated in self-learning and practical experimentation activities

Week 14 – NexusAgent and NexusCaller Experimental Development

- Worked on personal AI product projects including NexusAgent and NexusCaller
- Applied concepts learned from ERP, CRM, and voice AI development workflows
- Explored multi-agent communication and AI-assisted business workflow concepts
- Performed experimentation on conversational systems, routing logic, and AI integrations
- Worked on implementation ideas involving voice systems, automation, and AI agents
- Continued improving practical understanding of AI product development environments

Week 15 – Final Learning Reflection Workflow Consolidation and Continuation

- Reviewed overall internship learning experience and technical progression
- Consolidated understanding of AI systems, ERP workflows, CRM implementations, and development practices involved in building products
- Reflected on real-world project exposure, collaborative development, and implementation methodologies
- Improved understanding of debugging, testing, integration, and workflow optimization practices
- Strengthened practical knowledge in AI product development and software engineering workflows
- Gained confidence in handling real-time development tasks and professional engineering environments

3.4 ERP SYSTEM DEVELOPMENT EXPERIENCE

During the internship period, significant exposure was gained in the development and enhancement of the Rooman ERP system. The ERP system served as a real-time operational platform used for managing organizational workflows, student lifecycle processes, internal management activities, and administrative operations across different departments.

The development activities involved understanding the existing system architecture, module relationships, backend workflows, frontend integrations, and role-based operational structures implemented within the ERP environment. Exposure to production-oriented systems helped in understanding practical software engineering workflows, collaborative implementation processes, and real-world project development methodologies.

One of the major areas of contribution during this phase involved working on referral-based workflows and ERP-related module enhancements. Activities included planning and implementing referral lifecycle workflows, public referral pages, QR-based referral systems, and frontend/backend integration processes. The implementation process also involved workflow validation, debugging activities, and testing under different operational conditions.

Further development activities included work related to question bank systems, testing modules, dashboard improvements, and UI refinement tasks. The implementation workflows involved structured CRUD operations, module-level integrations, validation logic handling, role-based workflow understanding, and collaborative debugging processes. Additional activities included improving interface consistency, workflow navigation, and operational usability within different ERP modules.

The internship also provided practical exposure to frontend and backend integration workflows, API communication, system architecture understanding, and collaborative software development practices followed within production-oriented engineering environments. Debugging, testing, issue tracking, and implementation refinement activities were continuously carried out to improve system stability and workflow efficiency.

The ERP development phase contributed significantly toward improving practical understanding of

full-stack development workflows, production-level practices, operational system management, and collaborative engineering methodologies used in modern software development environments.

3.5 CRM SYSTEM DEVELOPMENT EXPERIENCE

During the internship period, exposure was also gained in the development and enhancement of the Rooman CRM system. The CRM system was designed to manage leads, communication workflows, customer interactions, operational activities, and business-related processes within the organization. Working on the CRM system provided practical understanding of workflow-driven application development and collaborative implementation methodologies used in real-world operational environments.

The development activities involved understanding CRM architecture, module relationships, backend workflows, frontend integrations, and operational lifecycle management systems. Exposure to real-time business workflows helped in understanding how lead management systems, communication modules, and workflow automation processes are implemented within production-oriented software environments.

Major implementation activities included work related to leads management workflows, calls modules, reusable component systems, filtering mechanisms, and analytics-oriented functionalities. The development process involved frontend and backend coordination, API integration workflows, operational data handling, and implementation of structured workflow logic across CRM modules.

Further activities included improving workflow consistency, enhancing interface usability, implementing reusable charts and filtering systems, debugging operational issues, and optimizing frontend/backend synchronization processes. Practical exposure was also gained in collaborative implementation activities, workflow-based development approaches, and system integration practices followed within CRM environments.

The CRM development phase provided hands-on experience with production-level workflow handling, operational module implementation, debugging methodologies, collaborative development environments, and modern software engineering practices. It also improved practical understanding of scalable workflow systems, integration handling, and real-world CRM-based application development processes.

3.6 FULL STACK DEVELOPMENT AND INTEGRATION ACTIVITIES

The internship provided practical exposure to full stack development workflows involving frontend implementation, backend development, database interactions, API communication, and system integration activities. Working on real-time ERP and CRM environments helped in understanding how different components within large-scale applications interact and operate together in production-oriented systems.

The development activities involved participation in frontend and backend integration workflows, implementation of operational features, API-based communication systems, structured data handling, and workflow synchronization processes. Exposure to both client-side and server-side development activities contributed toward improving understanding of real-world software engineering practices and collaborative implementation methodologies.

Frontend-related activities included interface enhancements, dashboard improvements, workflow-based UI refinements, responsive implementation tasks, usability improvements, and operational workflow handling across different modules. Backend-related activities involved workflow logic implementation, API integration, structured data processing, module coordination, validation handling, and operational functionality management.

The internship also involved working with integration workflows related to referral systems, QR-based processes, communication workflows, analytics-oriented features, and operational synchronization between multiple modules. Continuous debugging, testing, validation, and refinement activities were carried out to ensure workflow consistency and stable system behavior across implementation environments.

Exposure to collaborative development practices, production-level implementation workflows, and real-time operational systems significantly improved practical understanding of full stack development processes. The implementation activities helped in strengthening analytical thinking, debugging approaches, integration handling capabilities, and overall understanding of scalable software development environments.

3.7 DEBUGGING, TESTING, AND PROBLEM-SOLVING EXPERIENCE

Throughout the internship period, continuous debugging, testing, and problem-solving activities were

carried out across AI systems, ERP workflows, CRM modules, and integration environments. These activities played an important role in understanding real-world software engineering challenges and improving practical implementation capabilities within production-oriented systems.

The debugging activities involved identifying workflow issues, analysing system behaviour, testing operational functionalities, and resolving implementation-related problems across frontend and backend environments. Exposure to real-time development systems helped in understanding how debugging and validation processes are performed in collaborative engineering workflows.

In AI voice agent experimentation, multiple testing scenarios were carried out to analyse response latency, conversational continuity, model behaviour, routing workflows, and synchronization handling within real-time communication systems. Different models and frameworks behaved differently under various operational conditions, requiring continuous experimentation, testing, and optimization to improve workflow stability and conversational performance.

Within ERP and CRM development activities, debugging tasks included fixing workflow inconsistencies, resolving integration-related issues, improving validation mechanisms, refining operational logic, and enhancing module-level interactions. Testing activities were continuously performed to verify frontend/backend communication, API behaviour, workflow synchronization, and usability improvements across different modules and implementation environments.

The internship also involved practical problem-solving activities related to system integration, operational workflows, module coordination, and implementation challenges encountered during development tasks. These experiences improved analytical thinking, debugging methodologies, workflow analysis capabilities, and practical understanding of real-world engineering problem-solving approaches.

Overall, the debugging and testing activities contributed significantly toward improving technical confidence, implementation accuracy, troubleshooting abilities, and practical understanding of production-level software development workflows.

3.8 TOOLS AND TECHNOLOGIES APPLIED

During the internship period, various modern tools, frameworks, platforms, and technologies were used across AI research, voice agent development, ERP and CRM systems, full stack development,

and real-time communication workflows. Exposure to these technologies helped in understanding practical implementation methodologies, collaborative development environments, AI-assisted systems, and production-oriented software engineering practices followed within real-world development environments.

Table 3.8.1: Tools and Technologies Used During the Internship

Category	Tool / Technology	Purpose / Usage
Frontend Development	React, TypeScript, HTML, CSS	Building responsive user interfaces and frontend workflows
Backend Development	Python, Django, Django REST Framework, FastAPI	Developing backend logic, APIs, and server-side workflows
Database Management	PostgreSQL, SQLite, Convex	Managing structured application data and real-time workflows
Vector Database	ChromaDB	Storing embeddings and supporting RAG-based AI retrieval workflows
AI Frameworks and Models	Ollama, Claude, Codex, Qwen Coder	AI-assisted development, code generation, and model experimentation
AI Platforms	AWS Bedrock, NVIDIA NIM	Accessing and experimenting with foundation AI models and inference services
AI Orchestration	LangChain, RAG Pipelines	Building retrieval-augmented AI systems and orchestration workflows
NLP and PII Extraction	spaCy, Flair, HuggingFace Transformers, GLiNER, Microsoft Precidio	NLP experimentation, entity recognition, and PII extraction
Voice AI Frameworks	LiveKit, Pipecat, TEN Framework	Developing real-time AI voice communication systems
Speech Processing	Deepgram, ElevenLabs, Whisper, Kokoro TTS	Speech-to-Text (STT) and Text-to-Speech (TTS) processing
Real-Time Communication	WebRTC, WebSocket Streaming	Handling low-latency real-time communication and audio streaming
Communication APIs	Twilio, Telnyx, Wati	Voice communication and telephony integration workflow
Cloud and Infrastructure	AWS EC2, AWS S3	Cloud hosting, deployment, and storage management
API Integration	REST APIs, Webhooks	Connecting frontend, backend, ERP, CRM, and AI systems
Version Control	Git, GitHub	Source code management and collaboration
Development Environment	Visual Studio Code, Claude Code	Writing, debugging, and managing source code
Debugging and Testing	Chrome DevTools, Postman	Debugging applications and testing APIs/workflows

3.9 PERSONAL LEARNING AND PRODUCT DEVELOPMENT

Apart from the organizational projects and assigned development activities, the internship period also encouraged continuous self-learning, experimentation, and independent product development activities. The practical exposure gained through AI systems, ERP workflows, CRM development, and voice agent experimentation was further applied in personal research and development projects focused on AI-assisted workflows and conversational systems.

As part of continuous learning and practical implementation, experimental AI product projects such as NexusAgent and NexusCaller were developed to explore advanced conversational AI workflows, multi-agent communication systems, AI-assisted automation, and voice-based interaction models. These projects were inspired by the real-world development practices, AI architectures, and workflow methodologies experienced during the internship period.

Additional experimentation activities involved research and implementation related to Retrieval-Augmented Generation (RAG) workflows, AI orchestration systems, conversational memory handling, real-time communication pipelines, streaming-based voice systems, and AI-assisted business automation concepts. Practical exposure to modern AI tools, frameworks, and development environments helped in improving understanding of scalable AI product engineering and real-world implementation workflows.

The personal development activities carried out during the internship period contributed significantly toward improving independent problem-solving abilities, experimentation methodologies, technical confidence, and practical understanding of AI product development environments. These activities also helped in strengthening interest toward building scalable AI-driven systems and exploring modern approaches in conversational AI and intelligent workflow automation.

3.10 Overall Learning Reflection

The internship at Rooman Technologies provided valuable practical exposure to real-world software development environments, AI product engineering workflows, and collaborative implementation practices followed within modern technology organizations. The internship experience helped bridge the gap between academic learning and industry-oriented development processes through participation in live projects, research activities, and production-oriented implementation tasks.

Throughout the internship period, exposure was gained in multiple technical domains including Artificial Intelligence, conversational AI systems, NLP workflows, AI voice agents, ERP and CRM development, frontend and backend integration, debugging methodologies, and collaborative software engineering practices. Working on real-time systems and operational workflows significantly improved practical understanding of implementation strategies, workflow coordination, testing approaches, and scalable software development environments.

The internship also helped in improving analytical thinking, problem-solving abilities, debugging skills, communication, teamwork, and independent learning capabilities. Continuous experimentation with emerging technologies, AI frameworks, real-time communication systems, and workflow automation concepts contributed toward building technical confidence and a stronger understanding of modern AI-assisted development ecosystems.

One of the most valuable aspects of the internship was the opportunity to work on live projects and practical engineering challenges instead of isolated academic exercises. Exposure to real-world implementation workflows, collaborative development environments, and production-oriented systems helped in understanding professional software engineering practices and the responsibilities involved in product development environments.

Overall, the internship experience contributed significantly toward professional growth, technical improvement, and career readiness in the field of Artificial Intelligence, full stack development, and AI product engineering.

CHAPTER 4

LEARNING OUTCOMES AND PROJECT DETAILS

4.1 PROJECT OVERVIEW

During the internship at Rooman Technologies Pvt. Ltd., practical exposure was gained through involvement in multiple real-world projects related to Artificial Intelligence systems, ERP and CRM development, AI voice agents, and full stack application development. The internship focused on both research-based experimentation and production-oriented implementation activities within live organizational systems.

A major part of the internship involved working on AI voice agent systems and multi-agent conversational workflows using technologies related to Speech-to-Text (STT), Text-to-Speech (TTS), Large Language Models (LLMs), and real-time communication frameworks. Various AI models and orchestration workflows were tested to understand conversational continuity, routing systems, streaming workflows, and latency optimization in real-time environments.

The internship also included contributions to the Rooman ERP system through UI enhancements, workflow improvements, module development, bug fixing, QR-based workflows, question bank systems, and testing-related functionalities. In addition, work was carried out on CRM workflows including leads management, calls modules, analytics dashboards, reusable chart systems, and advanced smart filtering mechanisms.

Another important project area involved the development of a referral-based SaaS workflow integrated with ERP operations for agent referral management and QR-based onboarding processes. Along with organizational projects, independent AI product experimentation activities such as NexusAgent and NexusCaller were also explored based on the practical learning gained during the internship.

Overall, the internship provided valuable exposure to AI engineering, real-time communication systems, enterprise application development, debugging workflows, and collaborative software development practices in a production-oriented environment.

4.2 AI VOICE AGENT AND MULTI-AGENT SYSTEM

One of the major areas of work during the internship involved research, experimentation, and implementation related to AI voice agents and multi-agent conversational systems. The project focused on building real-time conversational workflows using Speech-to-Text (STT), Large Language Models (LLMs), and Text-to-Speech (TTS) technologies within a continuous communication environment.

The development activities included experimentation with conversational routing, persona switching, streaming workflows, latency optimization, and real-time audio communication systems. Multiple AI models, orchestration frameworks, and voice processing technologies were tested to understand conversational continuity and multi-agent interaction workflows in live environments.

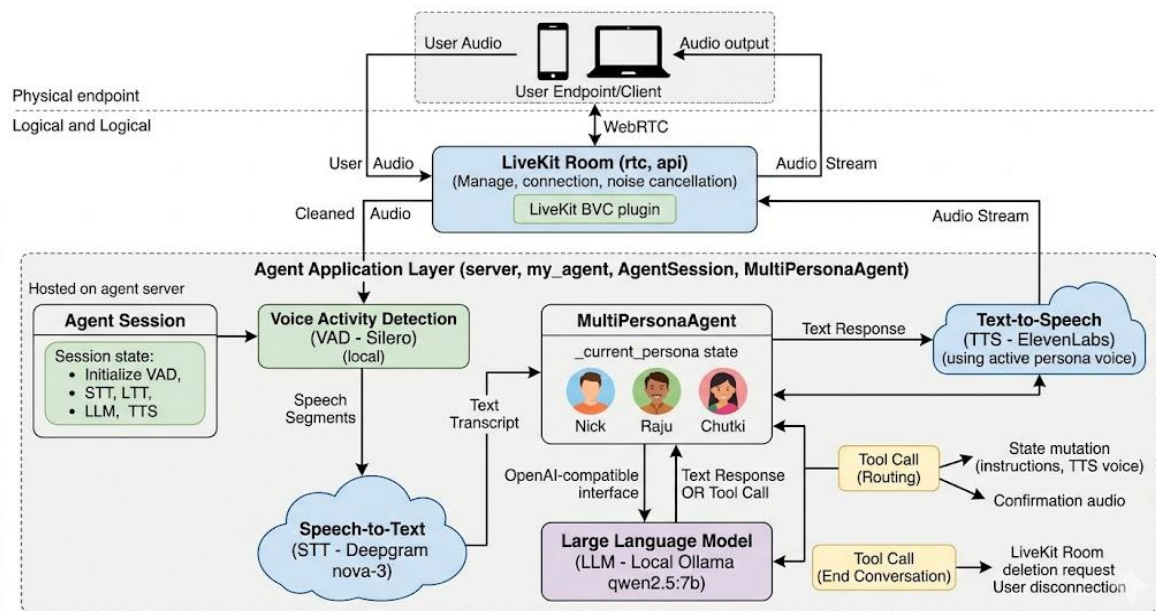


Figure 4.2.1: System Architecture of Multi-Agent Voice System

Multiple technologies such as LiveKit, Pipecat, TEN Framework, Ollama, Deepgram, Whisper, ElevenLabs, and Kokoro TTS were explored during the experimentation process. Different Large Language Models (LLMs) and open-source AI solutions were evaluated to analyze their performance under various conversational conditions including response latency, synchronization handling, routing logic, and streaming behavior.

The system was designed to support multi-agent orchestration workflows where multiple AI personas dynamically interact based on user intent and conversational flow. Testing activities were carried out

to understand persona switching, voice continuity, interruption handling, and low-latency communication workflows within real-time AI voice systems.

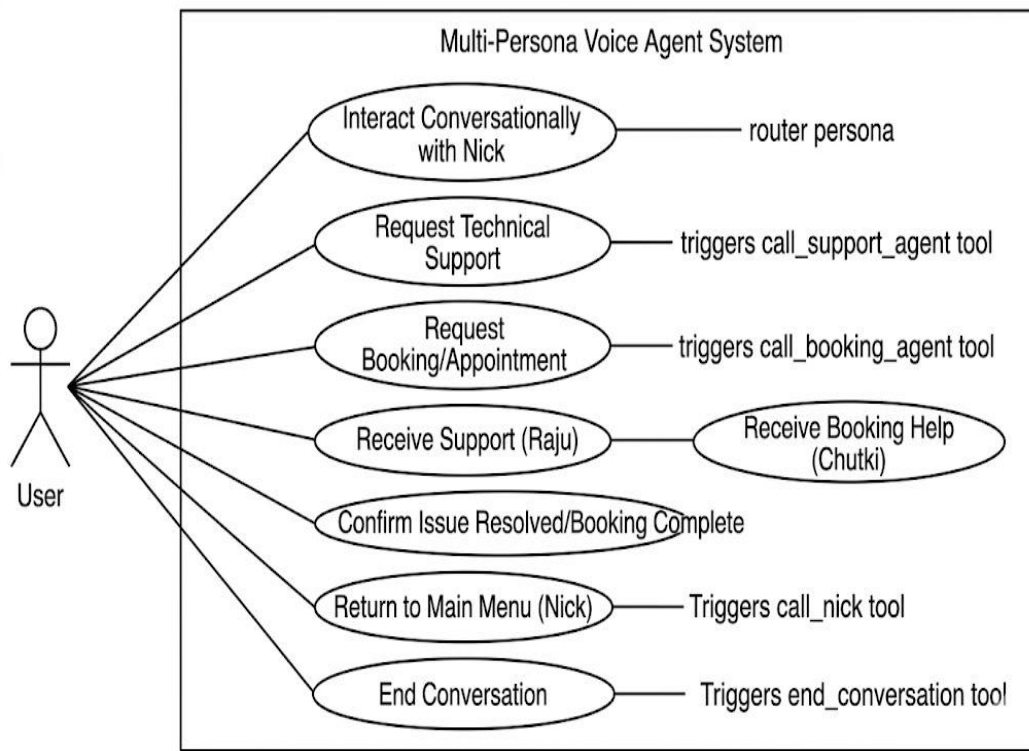


Figure 4.2.2: Use Case Diagram of Multi-Agent Workflow

The project also involved experimentation with conversational pipelines capable of handling routing workflows, contextual response generation, and real-time audio streaming using WebRTC-based communication systems. Additional research was conducted on open-source alternatives and AI orchestration workflows to improve flexibility, reduce dependency on commercial APIs, and explore scalable approaches for conversational AI development.

Workflow experimentation included integration testing, streaming analysis, debugging activities, and comparative evaluation of different STT, TTS, and LLM models based on real-time performance conditions. These activities significantly improved practical understanding of conversational AI engineering and production-oriented voice communication systems.

4.3 CRM VOICE AGENT SYSTEM

The CRM Voice Agent System was developed as an AI-assisted conversational workflow designed to support communication and interaction processes within CRM-related operations. The primary objective of this system was to explore how AI voice agents could be integrated with CRM workflows for handling conversational interactions, routing logic, and automated communication processes in real-time environments.

The development activities involved experimentation with AI-powered calling workflows, conversational routing systems, and multi-persona interaction handling within CRM-based communication environments. The system was designed to support intelligent conversational flows where AI agents could dynamically respond based on user intent and operational workflow conditions.

Different AI technologies and communication frameworks were integrated to analyze conversational continuity, response latency, streaming behavior, and real-time communication handling during voice interactions. The experimentation process also included testing conversational pipelines capable of supporting lead interaction workflows, automated communication handling, and AI-assisted routing systems for CRM operations.

The CRM Voice Agent system additionally involved experimentation with multi-agent conversational workflows where different AI personas could interact dynamically depending on the conversation flow and operational requirements. Testing activities were continuously carried out to analyze routing behavior, conversational continuity, synchronization handling, and workflow stability within real-time voice communication systems.

The practical implementation and experimentation carried out during this project provided valuable understanding of AI-assisted CRM communication systems, conversational workflow automation, and real-time AI voice interaction environments used in modern intelligent communication platforms.

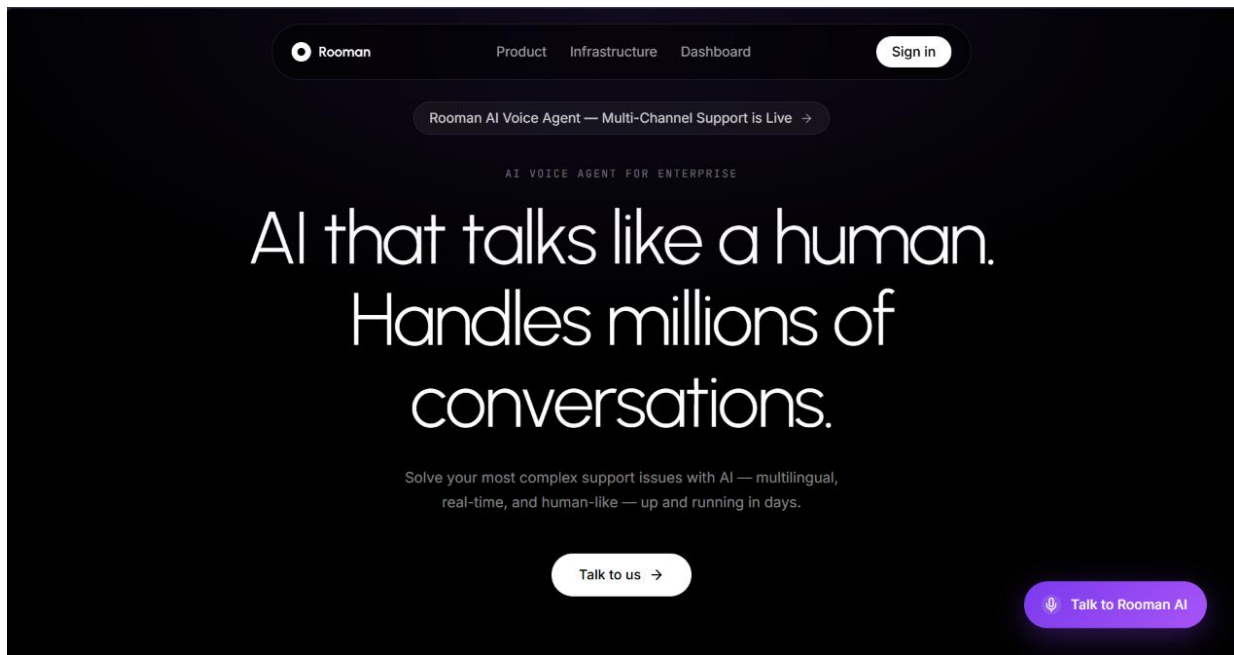


Figure 4.3.1: Voice CRM Landing Page

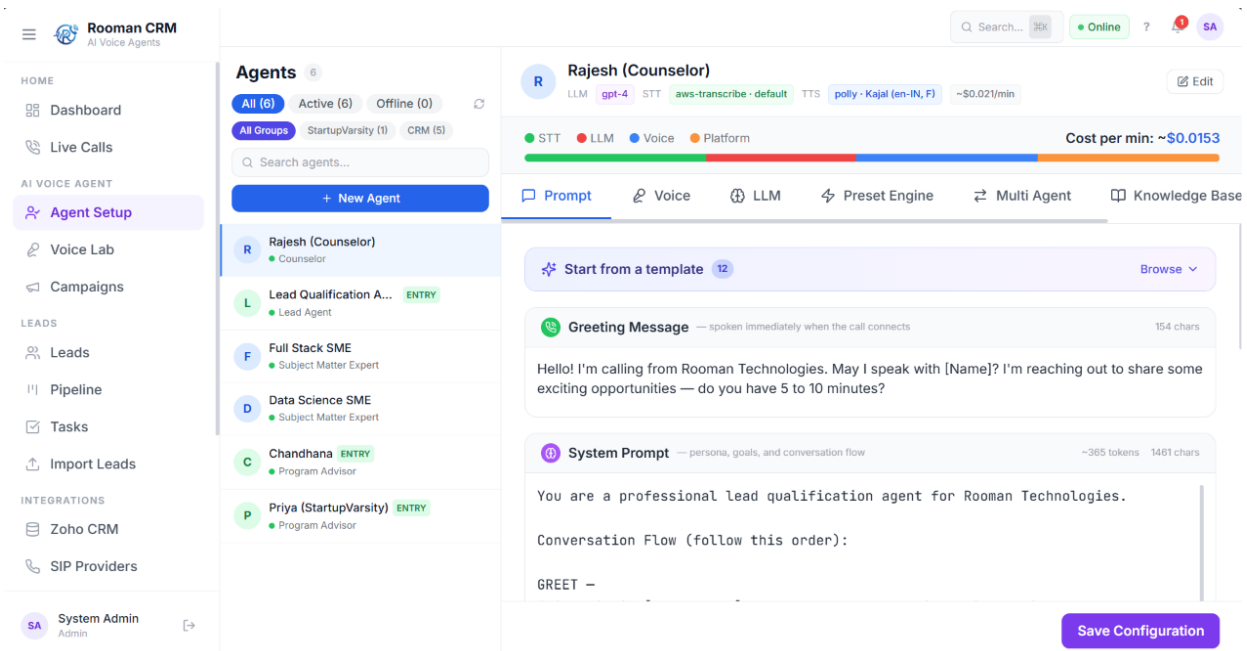


Figure 4.3.2: Agent Setup Page & Other Modules

4.4 ERP SYSTEM DEVELOPMENT AND WORKFLOW ENHANCEMENTS

A significant portion of the internship involved working on the Rooman ERP system, which is used for managing various operational workflows within the organization. The development activities included frontend enhancements, module development, workflow optimization, UI improvements, debugging activities, and implementation of operational functionalities across different ERP modules.

The work carried out during the internship involved contributions to multiple sections of the ERP system including landing pages, signup workflows, sidebar navigation, headers, breadcrumbs, dashboard interfaces, question bank systems, testing modules, and QR-based operational workflows. Practical exposure to a production-oriented ERP environment helped in understanding real-world workflow implementation, frontend/backend coordination, and collaborative software development practices.

One of the major areas of contribution involved improving user interface workflows and enhancing operational usability across different ERP modules. Activities included responsive UI improvements, workflow optimization, component restructuring, navigation handling, and bug fixing to improve the overall user experience and operational consistency within the system.

The internship also involved working on the Question Bank Module and Test Module within the ERP environment. Development activities included workflow implementation, module enhancements, CRUD-related operations, testing workflows, and AI-assisted proctoring-related functionalities involving violation detection concepts using face detection models.

Additional implementation activities involved QR-based workflow integration and operational process handling within ERP-related systems. Continuous debugging, testing, and workflow validation activities were carried out to ensure stable system behavior and smooth operational functionality across different modules.

Working on the ERP system provided valuable practical exposure to production-level application development, operational workflow handling, collaborative implementation methodologies, debugging approaches, and full stack development practices followed within real-world enterprise systems.

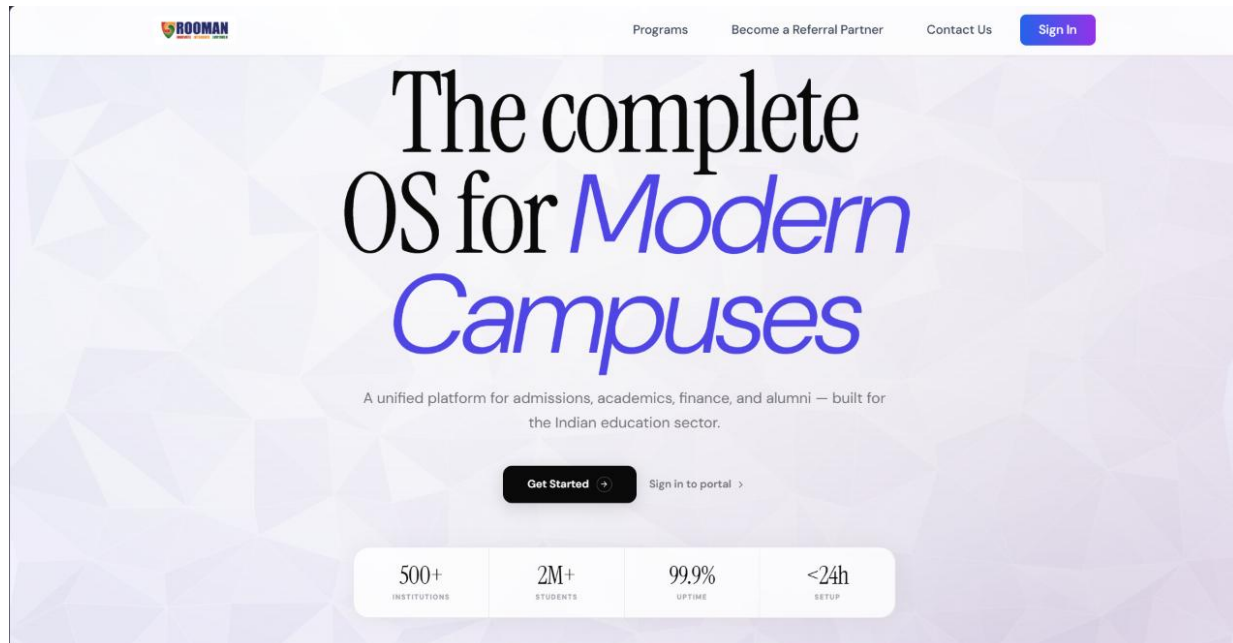


Figure 4.4.1: ERP Landing Page

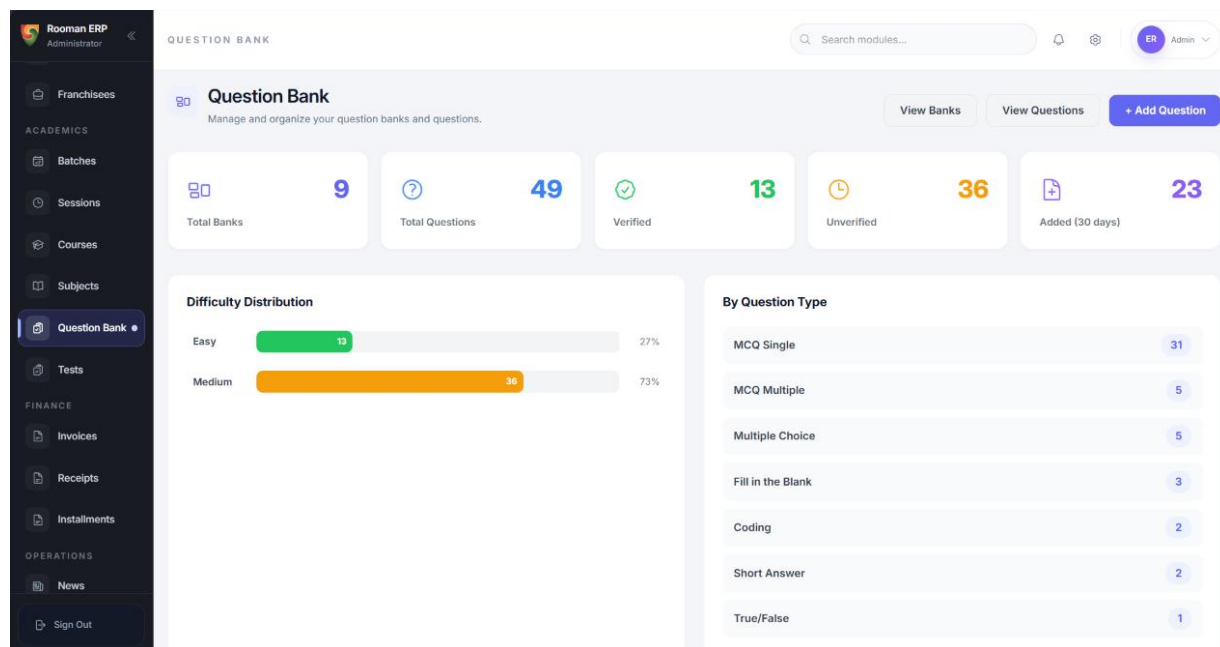


Figure 4.4.2: ERP Question Bank Module

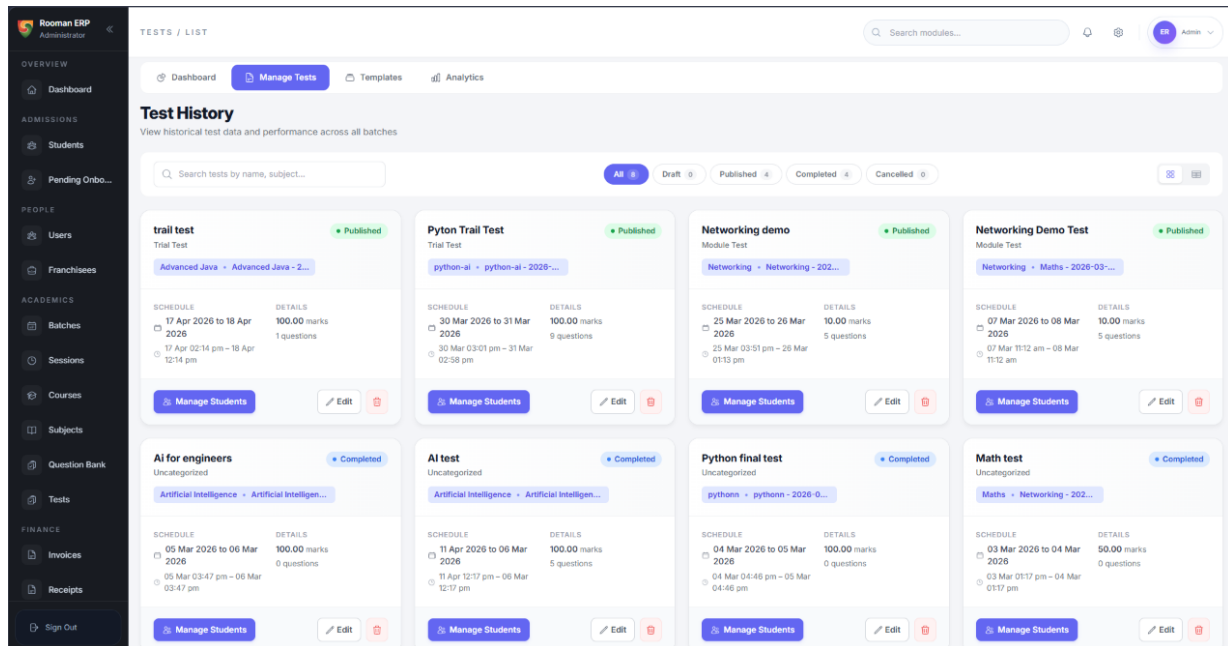


Figure 4.4.3: ERP Test Module

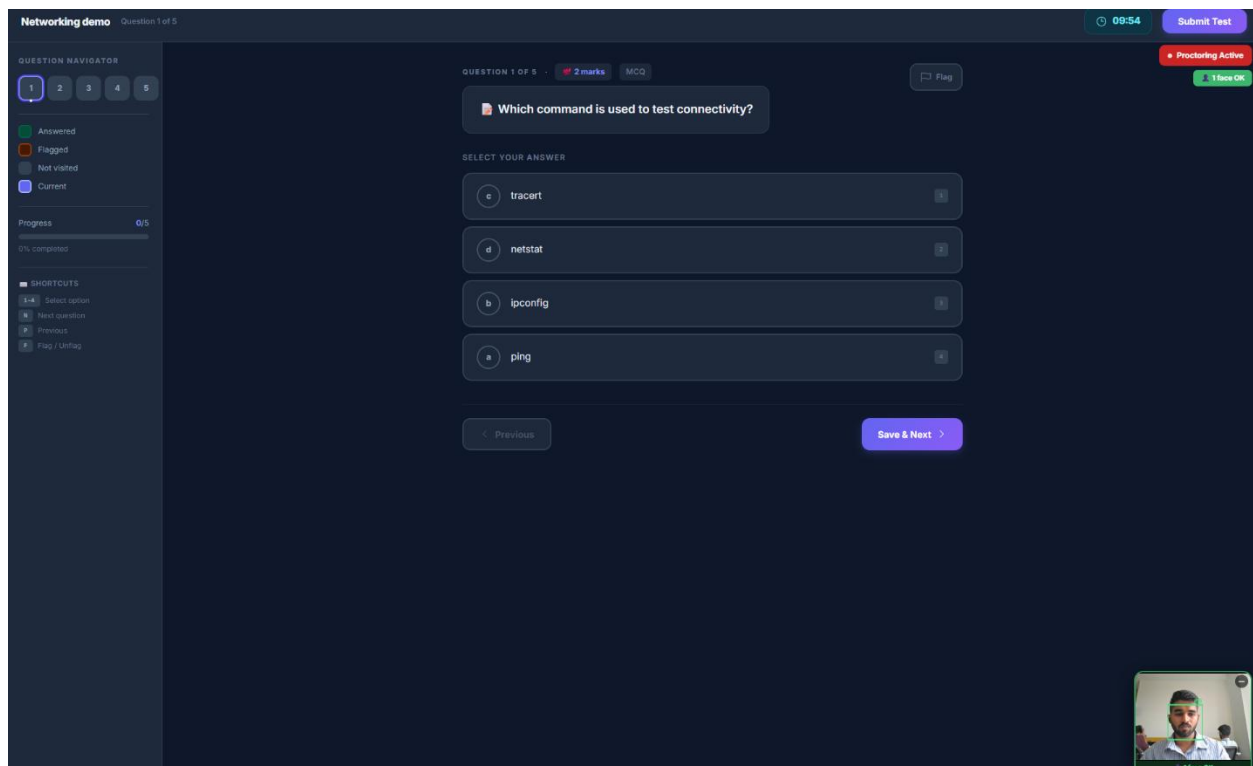


Figure 4.4.4: ERP Student Test Interface

Table 4.4.1: ERP Modules and Work Performed

Module	Work Performed
Landing Page	UI enhancement and responsive design improvements
Signup Workflow	Workflow optimization and validation improvements
Public Page	For leads enquiry and submit form
Sidebar and Header	Navigation and interface enhancements
Question Bank Module	CRUD workflows and operational implementation
Test Module	Testing workflows and AI proctoring-related functionalities
QR Workflow	QR-based operational workflow integration
Bug Fixing and UI Improvements	Workflow optimization and debugging activities

4.5 ERP REFERRAL SAAS SYSTEM

The ERP Referral SaaS System was developed as a referral management platform integrated with both the ERP and CRM environments to support referral onboarding, tracking, and operational management workflows. The system was designed for agents who could refer candidates, track referral activities, and manage the referral lifecycle through both web and mobile-based interfaces.

The project involved the development of referral workflows connected with ERP and CRM operations, enabling centralized management of referral-related activities within the organizational ecosystem. The platform included onboarding workflows, referral tracking systems, QR-based referral access, and operational workflow integration between multiple systems.

One of the important features of the system was the Flutter-based mobile application developed for agents to monitor their referral activities, onboarding status, referral progress, and operational workflows in real-time. The mobile application was designed to provide a user-friendly interface for agents to manage and track their complete referral process efficiently.

The project also included frontend and backend integration activities, API communication workflows, dashboard handling, QR-based referral systems, and workflow automation processes for referral management. Additional implementation activities involved debugging workflows, operational validation, and synchronization handling between ERP, CRM, and referral system environments.

Working on the ERP Referral SaaS System provided practical exposure to SaaS workflow development, multi-system integration, mobile application workflows, operational automation, and production-oriented implementation methodologies used in modern enterprise applications.

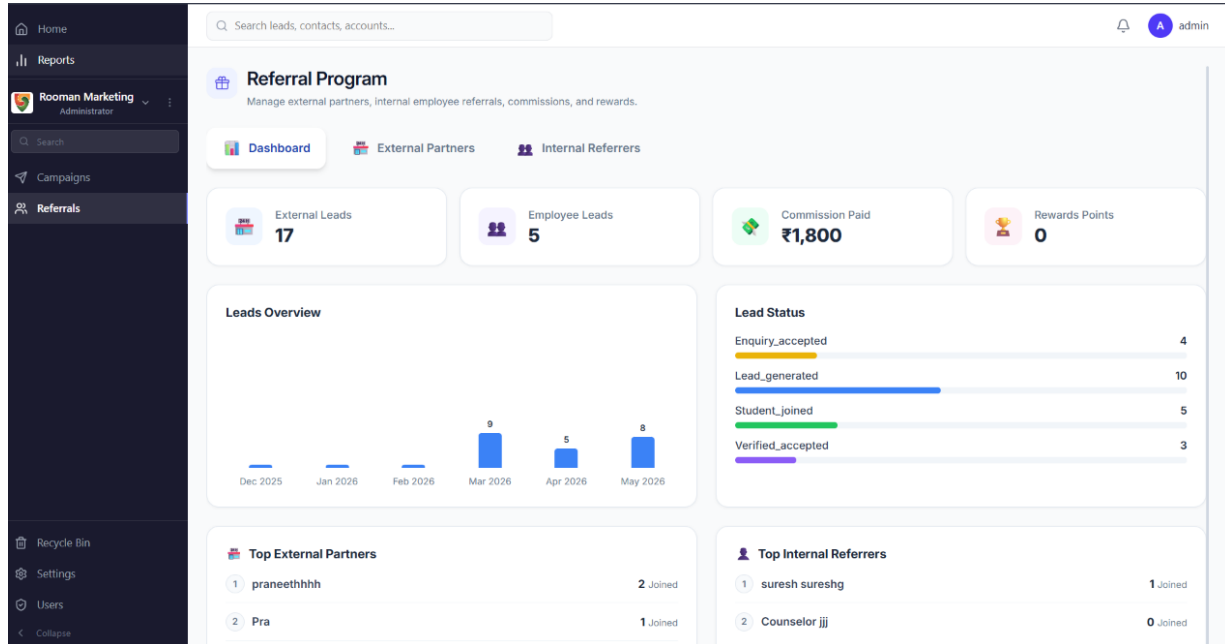


Figure 4.5.1: Referral Dashboard Page

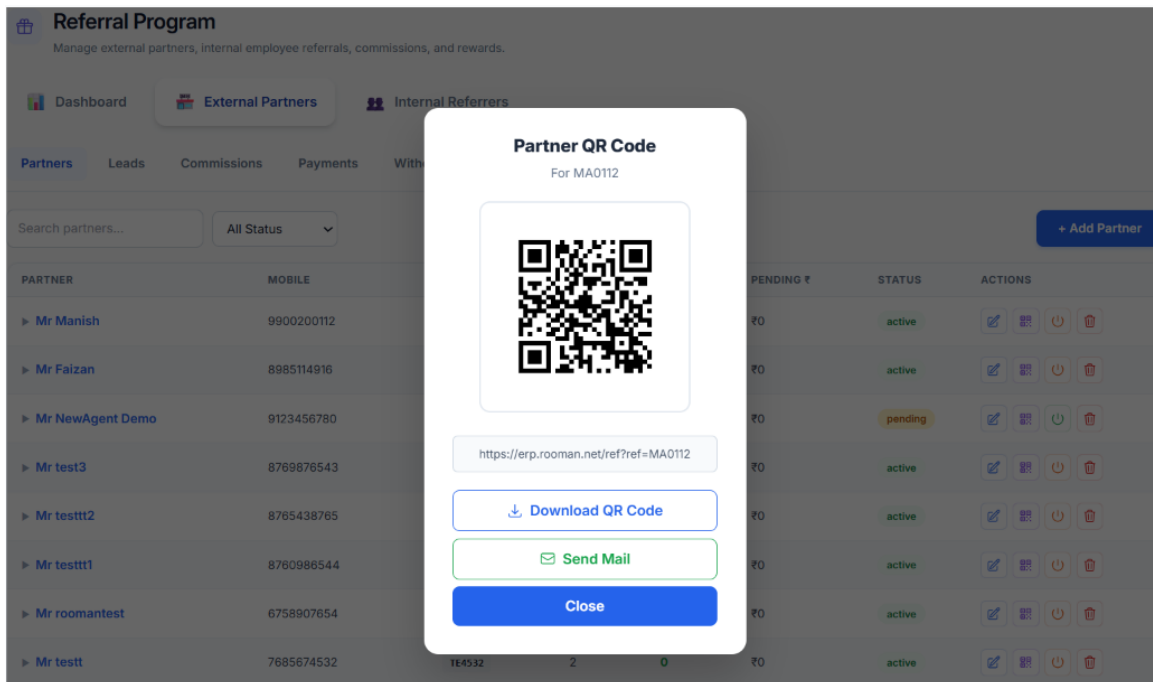


Figure 4.5.2: Referral QR Code Page

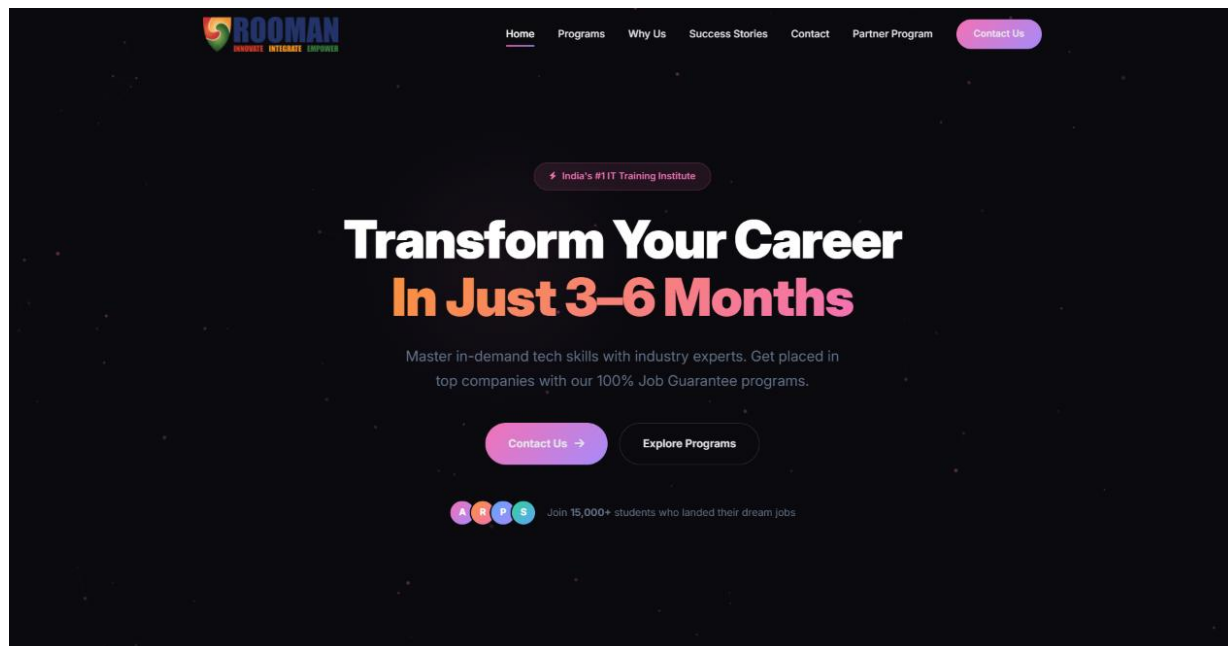


Figure 4.5.3: Public Page for Enquiry

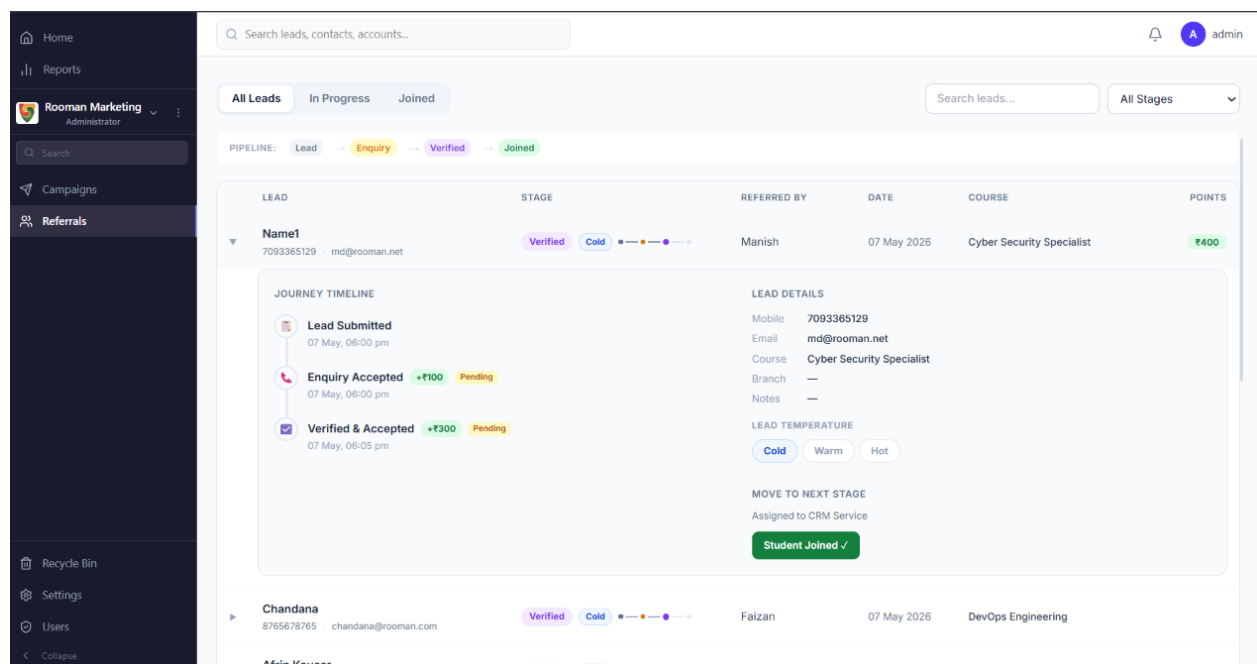


Figure 4.5.4: Lead Collection from QR or Referrals

Table 4.5.1: ERP Referral SaaS and Work Performed

Module	Description
Referral Registration	Agent onboarding and referral creation workflows
Referral Tracking	Monitoring referral progress and operational activities
QR-Based Workflow	QR-enabled onboarding and referral access
ERP Integration	Integration with ERP operational workflows
CRM Integration	Integration with CRM-based communication workflows
Flutter Mobile Application	Mobile interface for agents to manage referral activities
Dashboard and Analytics	Referral monitoring and workflow tracking

4.6 CRM SYSTEM DEVELOPMENT

The CRM system development activities focused on improving operational workflows related to lead management, communication tracking, analytics, and workflow optimization within the organizational CRM environment. The implementation process involved working on multiple CRM modules and enhancing various frontend and backend workflows used for managing operational activities and communication-related processes.

One of the major areas of contribution involved the Leads Module, where workflow handling, operational data management, and interface-related improvements were carried out to improve lead tracking and management processes. Additional work was also performed on the Calls Module involving communication workflow handling, operational coordination, and module-level improvements within CRM-related systems.

A significant part of the CRM development work involved the implementation and optimization of the Smart Filtering system, which supported advanced filtering workflows across operational CRM

data. The filtering system involved handling multiple filtering conditions, dynamic workflow logic, operational query management, and frontend/backend synchronization processes to support efficient data filtering and workflow management within the CRM environment.

The internship also involved implementation activities related to reusable charts, analytics workflows, dashboard improvements, and operational data visualization systems. Different workflow optimization and debugging activities were continuously carried out to improve module performance, operational usability, and frontend/backend coordination across CRM-related workflows.

Working on the CRM system provided practical understanding of workflow-driven application development, advanced filtering systems, operational data handling, collaborative engineering workflows, and production-oriented CRM implementation methodologies used in real-world enterprise applications.

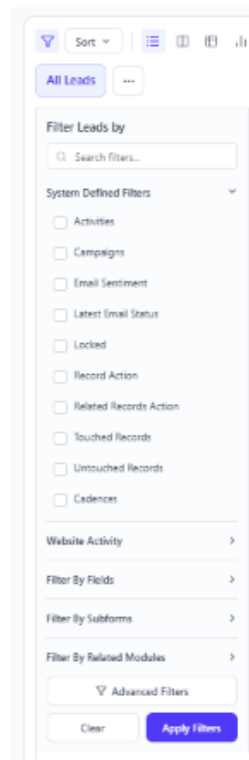


Figure 4.6.1: CRM Smart Filters

Leads
Capture - Quality - Convert

Search leads, contacts, accounts...

Sort | Filter | View | Refresh | Export

Viewing all leads | Import | More Tools | + Create Lead

Filter Leads by

Search filters...

System Defined Filters

- ☐ Activities
- ☐ Campaigns
- ☐ Email Sentiment
- ☐ Latest Email Status
- ☐ Locked
- ☐ Record Action
- ☐ Related Records Action
- ☐ Touched Records
- ☐ Untouched Records
- ☐ Cadences

Website Activity

Filter By Fields

	LEAD NAME	EMAIL	MOBILE	LEAD SOURCE	LEAD STATUS	ACCO
☐	CC Christina Chacko	christina@gmail.com	5675847987	Website	New	faroo
☐	NJ Nithya J	nitya@rooman.com	94859583932	Website	New	faroo
☐	MG Ms. Gagana R	gagana@rooman.com	9856834695	Referral	Enrolled	Unass
☐	AS Amit Sharma	amit.sharma@example.com	+919876543212	Referral	Qualified	faroo
☐	PP Priya Patel	priya.patel@example.com	+919876543211	Google Ads	Contacted	faroo
☐	RK Rahul Kumar	rahul.kumar@example.com	+919876543210	Website	New	khadi

Figure 4.6.2: CRM Leads Module Page

Rooman CRM
Administrator

Search leads, contacts, accounts...

Search calls

+ Create Call

Filter Calls by

Search filters...

All Types (22)

- Outbound 22
- Inbound 0
- Missed 0

System Defined Filters

- ☐ Record Action
- ☐ Related Records Action
- ☐ Touched Records
- ☐ Untouched Records

Filter By Fields

- ☐ Call Agenda
- ☐ Call Duration
- ☐ Call Duration (in seconds)
- ☐ Call Owner
- ☐ Call Purpose

	SUBJECT	CALL TYPE	CALL START TIME	DURATION	RESULT	STATUS	RELATED TO	CALL OWNER	ACTIONS
☐	CS Course selling	Outbound		01:00	not interested	completed			
☐	CT Call to Sahana Rao	Outbound	May 14, 2026, 04:53 PM	-		initiated		A admin@rooman.com	
☐	CT Call to sandeep1 S	Outbound	May 13, 2026, 01:20 PM	-		initiated		S Sandeep	
☐	CT Call to Afrin Kousar	Outbound	May 11, 2026, 04:49 PM	-		initiated		A admin@rooman.com	
☐	CT Call to Afrin Kousar	Outbound	May 11, 2026, 04:44 PM	-		initiated		A admin@rooman.com	
☐	CT Call to Afrin Kousar	Outbound	May 11, 2026, 04:42 PM	-		initiated		A admin@rooman.com	
☐	CT Call to Afrin Kousar	Outbound	May 11, 2026, 04:01 PM	-		initiated		A admin@rooman.com	

1-10 of 10 calls

< 1 / 1 >

Figure 4.6.3: CRM Calls Module Page

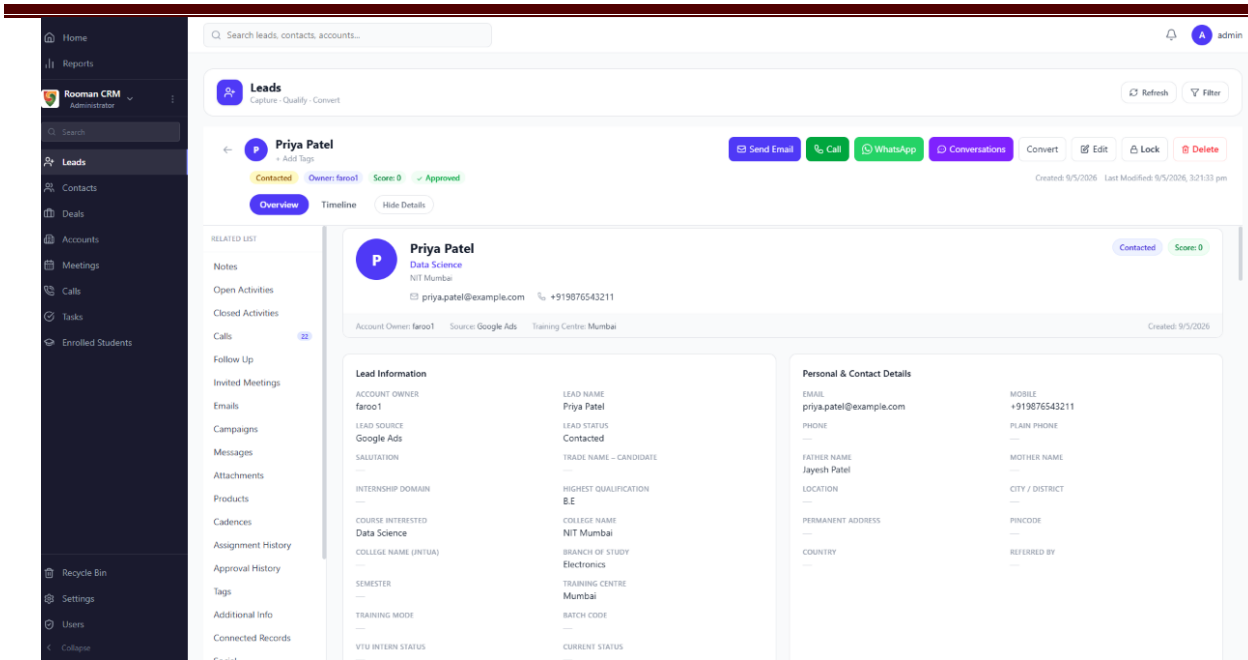


Figure 4.6.4: CRM Leads Profile Page

Table 4.6.1 – CRM Modules and Features

Module	Features Implemented
Leads Module	Lead management and workflow handling
Calls Module	Communication workflow management
Smart Filtering System	Advanced filtering and operational query handling
Charts and Analytics	Data visualization and reporting workflows
Dashboard Workflows	Operational monitoring and workflow optimization
Frontend/Backend Integration	Workflow synchronization and API integration

4.7 PERSONAL AI PRODUCT DEVELOPMENT AND RESEARCH

Apart from the organizational projects carried out during the internship, continuous independent experimentation and personal AI product development activities were also pursued based on the practical learning and technical exposure gained throughout the internship period. These activities focused on exploring scalable AI-assisted systems, conversational workflows, multi-agent orchestration environments, and real-time communication architectures.

One of the major personal experimentation projects involved the development of NexusAgent, an AI-assisted workflow system focused on conversational AI integration, automation workflows, and intelligent operational assistance. The project was developed as a learning-oriented AI product experimentation platform to explore real-world AI engineering concepts and scalable conversational system architectures.

Additional experimentation was carried out through NexusCaller and other multi-agent conversational AI projects involving real-time communication workflows, AI orchestration systems, streaming-based conversational pipelines, and multi-agent routing environments. These projects involved experimentation with Speech-to-Text (STT), Text-to-Speech (TTS), Large Language Models (LLMs), orchestration workflows, and real-time conversational systems using modern AI frameworks and communication technologies.

The personal development activities also included research and experimentation related to Retrieval-Augmented Generation (RAG), conversational memory handling, AI orchestration pipelines, real-time voice systems, and workflow automation concepts. Multiple open-source AI frameworks, communication systems, and orchestration technologies were explored to understand scalable approaches for AI-assisted product development.

These independent experimentation activities significantly contributed toward improving practical understanding of AI engineering, product architecture design, workflow automation, conversational systems, and production-oriented AI development methodologies beyond the organizational internship activities.

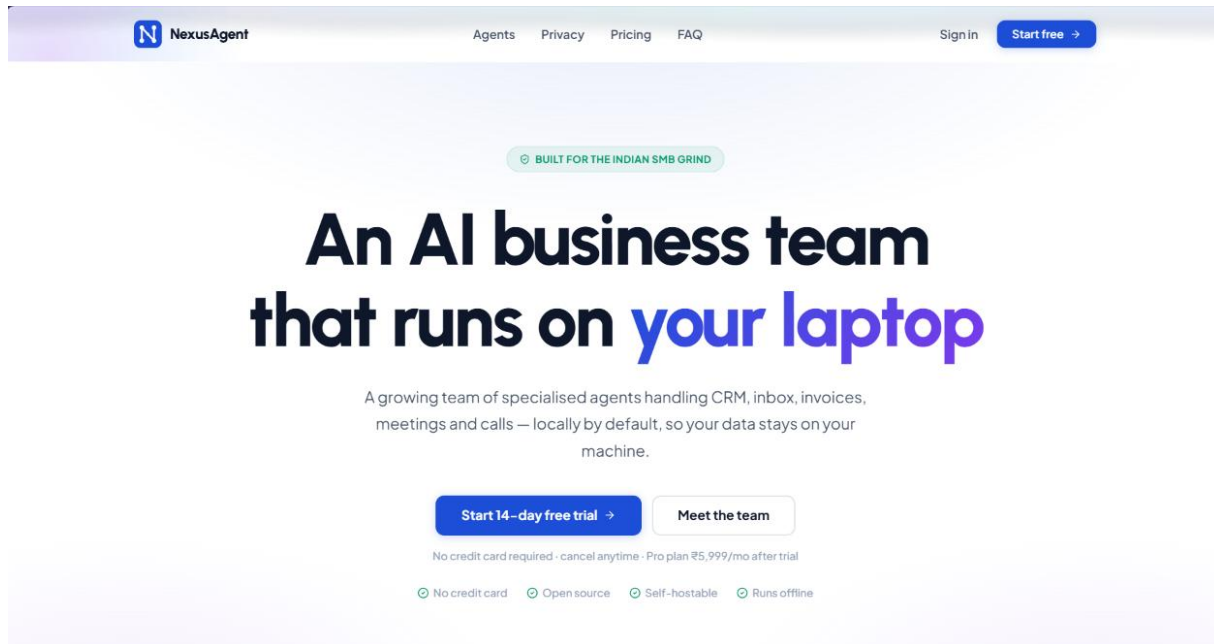


Figure 4.7.1: NexusAgent Landing Page

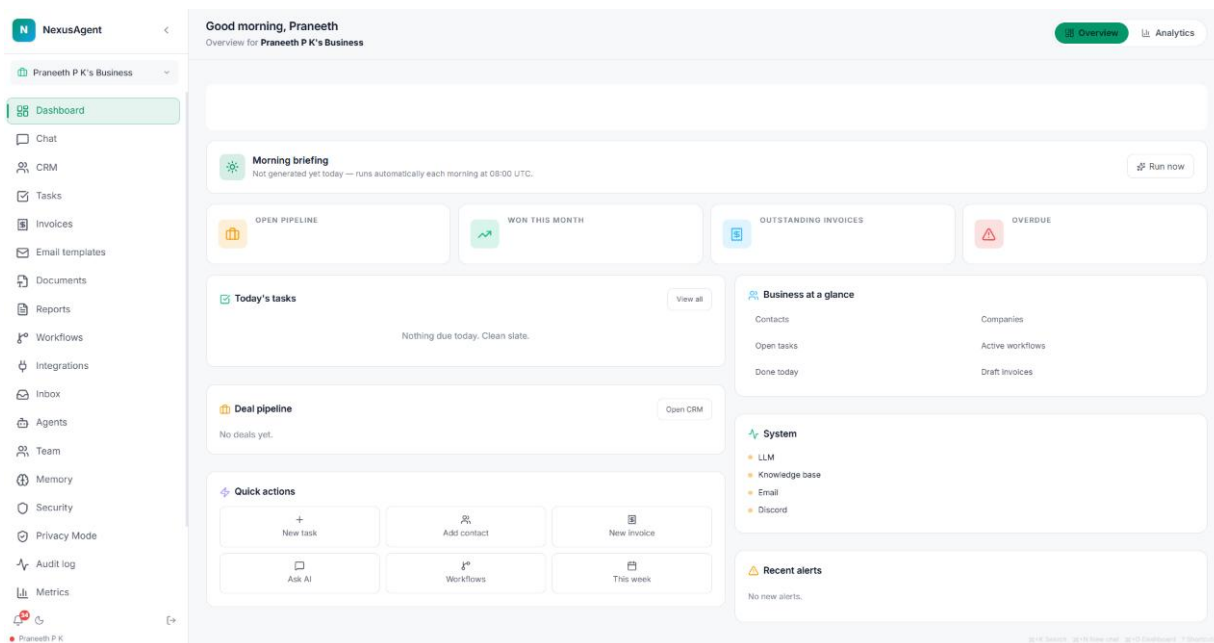


Figure 4.7.2: NexusAgent Dashboard

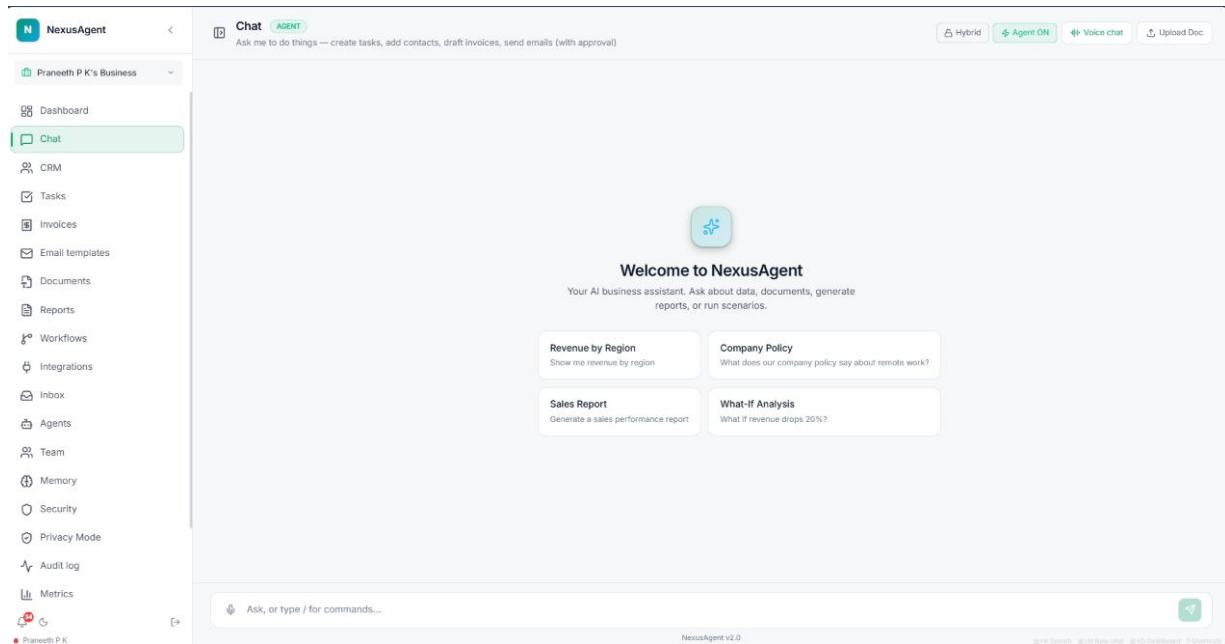


Figure 4.7.3: NexusAgent Chatbot Page

Table 4.7.1: Features of NexusAgent

Project / Research Area	Purpose
NexusAgent	AI-assisted workflow and conversational system experimentation
NexusCaller	Real-time AI voice communication workflows
Multi-Agent Systems	Multi-persona conversational orchestration
RAG Workflows	Retrieval-based AI response systems
AI Voice Systems	Real-time conversational AI experimentation
AI Orchestration Research	Workflow automation and agent coordination systems

4.8 TOOLS AND TECHNOLOGIES APPLIED

During the internship period, multiple modern technologies, frameworks, platforms, and development environments were used across AI systems, ERP development, CRM workflows, real-time communication systems, and full stack application development. Practical exposure to these technologies helped in understanding production-oriented software engineering workflows, AI-assisted systems, cloud-based environments, and collaborative development methodologies followed in real-world projects.

The implementation activities involved frontend development, backend development, conversational AI systems, speech processing workflows, AI orchestration pipelines, real-time communication systems, cloud infrastructure, API integration workflows, and debugging/testing environments. Different technologies were selected and applied based on project requirements, workflow complexity, scalability needs, and operational implementation conditions.

The major tools and technologies used during the internship are listed below.

Table 4.8.1: Tools and Technologies Applied

Category	Tool / Technology	Purpose / Usage
Frontend Development	React, TypeScript, HTML, CSS	Building responsive user interfaces and frontend workflows
Backend Development	Python, Django, Django REST Framework, FastAPI	Developing backend logic, APIs, and server-side workflows
Database Management	PostgreSQL, SQLite, Convex	Managing structured application data and real-time workflows

Vector Database	ChromaDB	Supporting RAG-based AI retrieval workflows
AI Frameworks and Models	Ollama, Claude, Codex, Qwen Coder	AI-assisted development and model experimentation
AI Platforms	AWS Bedrock, NVIDIA NIM	AI model access and inference services
AI Orchestration	LangChain, RAG Pipelines	AI workflow orchestration and retrieval systems
NLP and PII Extraction	spaCy, Flair, HuggingFace Transformers, GLiNER	NLP experimentation and PII extraction workflows
Voice AI Frameworks	LiveKit, Pipecat, TEN Framework	Real-time AI voice communication systems
Speech Processing	Deepgram, ElevenLabs, Whisper, Kokoro TTS	Speech recognition and voice synthesis
Real-Time Communication	WebRTC, WebSocket Streaming	Real-time audio and conversational streaming
Communication APIs	Twilio	Telephony and communication workflows
Cloud and Infrastructure	AWS EC2, AWS S3	Cloud hosting and storage management
API Integration	REST APIs, Webhooks	Connecting frontend, backend, ERP, CRM, and AI systems
Version Control	Git, GitHub	Source code management and collaboration
Development Environment	Visual Studio Code, Claude Code	Writing, debugging, and managing source code
Debugging and Testing	Chrome DevTools, Postman	API testing and workflow debugging

4.9 TESTING, DEBUGGING, AND IMPLEMENTATION CHALLENGES

Throughout the internship period, continuous testing, debugging, and workflow validation activities were carried out across AI systems, ERP modules, CRM workflows, and real-time communication environments. These activities played an important role in understanding production-level software engineering challenges and improving practical problem-solving capabilities within collaborative development environments.

One of the major challenges encountered during AI voice agent experimentation involved handling conversational continuity, response latency, synchronization issues, and routing behavior across multi-agent conversational systems. Different STT, TTS, and LLM models behaved differently under real-time conversational conditions, requiring continuous experimentation, comparative evaluation, and workflow optimization to improve conversational stability and response quality.

Additional debugging activities involved frontend/backend synchronization issues, API communication problems, operational workflow inconsistencies, UI-related bugs, and integration-related challenges across ERP and CRM environments. Workflow validation and testing activities were continuously performed to ensure stable functionality, smooth operational behavior, and reliable communication between multiple modules and systems.

The Smart Filtering workflow within the CRM system also involved complex implementation and debugging activities due to the handling of multiple dynamic filtering conditions and operational query management processes. Continuous testing and optimization were required to improve workflow efficiency, frontend/backend coordination, and operational usability within the CRM environment.

The internship also involved practical testing and validation activities related to AI orchestration workflows, real-time communication pipelines, QR-based workflows, referral systems, and operational automation processes. Different debugging approaches, workflow analysis methods, and iterative implementation strategies were used to identify and resolve technical issues encountered during development activities.

Overall, the testing and debugging processes carried out during the internship significantly improved

analytical thinking, troubleshooting methodologies, implementation accuracy, workflow validation practices, and practical understanding of production-oriented software engineering environments.

4.10 LEARNING OUTCOMES

The internship at Rooman Technologies Pvt. Ltd. provided valuable practical exposure to real-world software engineering environments, AI-assisted systems, enterprise application development, and production-oriented implementation workflows. Working on live projects and operational systems significantly improved both technical understanding and practical development capabilities across multiple domains.

The internship contributed toward improving knowledge in Artificial Intelligence systems, conversational AI workflows, real-time communication architectures, ERP and CRM development, frontend/backend integration, debugging methodologies, and workflow automation practices. Exposure to production-level systems helped in understanding how scalable applications, operational workflows, and collaborative engineering environments function in real-world organizations.

Practical implementation activities involving AI voice agents, multi-agent orchestration systems, smart filtering workflows, referral SaaS systems, and enterprise application modules helped strengthen analytical thinking, problem-solving abilities, debugging skills, and system-level understanding. Continuous experimentation with different frameworks, AI models, and communication technologies also improved independent learning capabilities and technical adaptability.

The internship additionally helped in developing professional skills related to teamwork, workflow coordination, project handling, collaborative implementation, and production-oriented development practices. Exposure to real operational systems and live project environments provided a clearer understanding of industry workflows, engineering responsibilities, and modern software development methodologies.

Overall, the internship experience significantly contributed toward professional growth, technical confidence, AI engineering exposure, and practical readiness for working in modern software product development and Artificial Intelligence domains.

CHAPTER 5

CONCLUSION

The internship at Rooman Technologies Pvt. Ltd. provided valuable practical exposure to real-world software development, Artificial Intelligence systems, enterprise application workflows, and production-oriented engineering practices. The internship experience helped bridge the gap between academic learning and industry implementation by providing opportunities to work on live projects involving AI voice agents, ERP systems, CRM workflows, referral SaaS platforms, and real-time communication systems.

Throughout the internship period, practical knowledge was gained in frontend and backend development, AI orchestration workflows, conversational AI systems, database management, API integration, debugging methodologies, and workflow automation processes. Working on multiple projects and operational systems helped improve technical understanding, analytical thinking, problem-solving abilities, and implementation skills within collaborative development environments.

The internship also provided significant exposure to modern AI technologies including Speech-to-Text (STT), Text-to-Speech (TTS), Large Language Models (LLMs), multi-agent orchestration systems, and real-time communication frameworks. Experimentation with different AI models, frameworks, and workflow architectures helped in understanding practical challenges related to conversational continuity, latency optimization, routing workflows, and production-level AI system implementation.

In addition to technical development, the internship contributed toward improving professional skills such as teamwork, workflow coordination, communication, debugging practices, and independent learning capabilities. Exposure to real-time organizational workflows and production-oriented systems provided a better understanding of industry standards, software engineering methodologies, and practical development environments.

Overall, the internship experience at Rooman Technologies Pvt. Ltd. played an important role in strengthening technical, practical skills, and enhancing future opportunities.

